

Unit 1 – General & Systematic Veterinary Bacteriology

All 34 syllabus topics compressed to six working sheets. Built so that one uninterrupted 30-minute pass gives you the diagram, the differentiating table, the one-liner and the answer skeleton for every examinable item in Paper I, Unit 1.

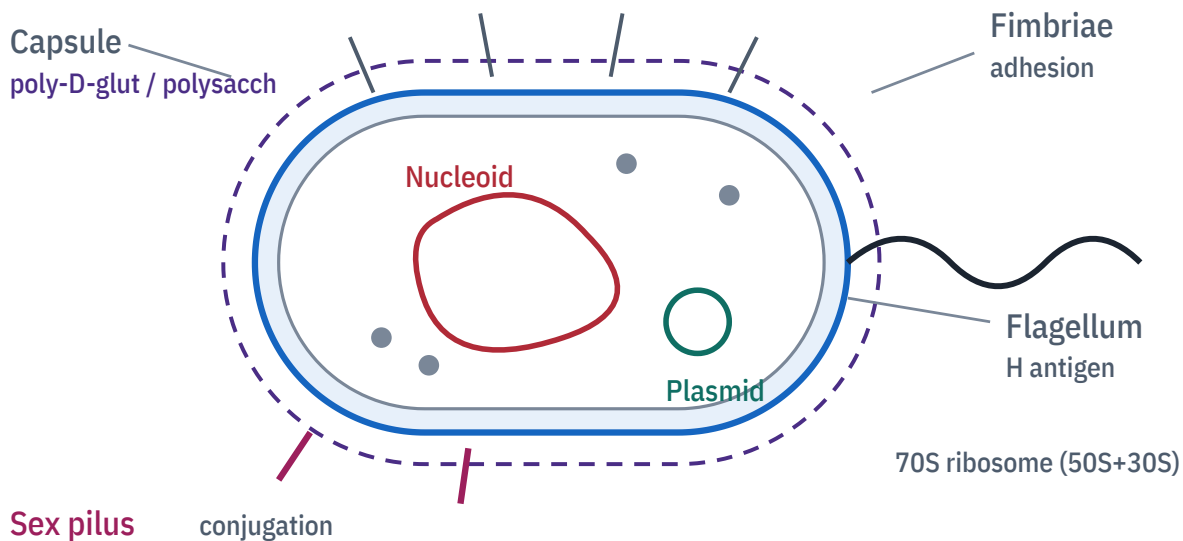
PAPER	SYLLABUS TOPICS	REVISION TIME	COURSE
Paper I · Units 1, 2, 3	34 of 34 covered	30 min full pass	B.V.Sc & A.H. II Yr
Theory 100 · Practical 60		12 min sheets 5–6 only	VCI MSVE · 3+2=5

Sheet	What it covers	Syllabus topics folded in
01	The bacterial cell — structure, staining, growth, microscopy	t01–t07
02	Infection, virulence, toxins, genetics, antibiotic resistance	t08–t12
03	Gram-positive cocci & non-sporing rods	t13–t16
04	Spore-formers, acid-fast & filamentous bacteria	t17–t20
05	Gram-negative rods — the whole enteric & respiratory block	t21–t28
06	Anaerobes, spirochaetes, cell-wall-deficient & obligate intracellular	t29–t34

■ General / structural
 ■ Gram-positive
 ■ Gram-negative
 ■ Acid-fast
 ■ Anaerobe / spore
■ Exam trap & mnemonic

Cell architecture

MECHANISM



- **Cell wall** = peptidoglycan (murein): NAM–NAG backbone, cross-linked by transpeptidase – the **penicillin target**. Absent in *Mycoplasma*.
- **Gram +ve**: 20–80 nm thick peptidoglycan (~50–90% of wall), **teichoic + lipoteichoic acid**, no outer membrane, no periplasm.
- **Gram –ve**: 2–7 nm peptidoglycan, **outer membrane with LPS**, periplasmic space, porins, Braun's lipoprotein.
- **LPS = Lipid A (toxic, endotoxin) + core + O-side chain (O antigen)**.
- Mesosome = infolding of plasma membrane; site of septum formation & DNA segregation.
- Spore layers outward: core → cortex (**Ca-dipicolinate**) → spore coat → exosporium.

Gram-positive vs Gram-negative

TABLE

Feature	Gram +ve	Gram –ve
Final colour	Violet	Pink / red
Peptidoglycan	Thick, multilayer	Thin, single layer
Teichoic acid	Present	Absent
Outer memb. / LPS	Absent	Present
Periplasm	Absent	Present
Toxin	Mostly exotoxin	Endotoxin ± exotoxin
Lysozyme	Sensitive	Resistant (OM barrier)
Penicillin	Sensitive	Relatively resistant
Wall-less form	Protoplast	Spheroplast

Feature	Gram +ve	Gram -ve
Flagellar rings	2 (S, M)	4 (L, P, S, M)

Trap: Gram reaction is a *wall* property, not a taxonomic one — old, autolysed or antibiotic-treated Gram +ve cells stain Gram-variable. *Mycobacterium* is technically Gram +ve but does not take Gram stain.

Staining — what each step actually does

MECHANISM

1 · Crystal violet

60 s — primary stain

2 · Gram's iodine

mordant → CV-I complex

3 · Decolouriser

acetone-alcohol, 5–10 s

4 · Safranin

counterstain, 30 s

Gram POSITIVE

Thick peptidoglycan dehydrates, pores shut → CV-I trapped → stays VIOLET

Gram NEGATIVE

Alcohol dissolves the lipid-rich outer membrane → CV-I washes out → takes PINK

Single most common mistake: over-decolourisation. Beyond ~10 s even true Gram +ve cells go pink. Decolouriser is the *only* timed-to-the-second step.

Stains you must be able to name

RAPID-FIRE

Simple	Methylene blue / basic fuchsin — shape & arrangement only
Negative	India ink, nigrosin — <i>Cryptococcus</i> capsule; background stained, cell clear
Gram (1884)	Christian Gram — differential; violet vs pink
Ziehl-Neelsen	Carbol fuchsin + heat, 20% H ₂ SO ₄ + alcohol, methylene blue → AFB red on blue
Modified ZN	0.5–1% H ₂ SO ₄ only — <i>Brucella</i> , <i>Nocardia</i> , <i>Coxiella</i> , <i>Chlamydia</i> (partially acid-fast)
Capsule	McFadyean's (polychrome methylene blue) — blue bacilli, pink capsule ; Anthony's (CuSO ₄)
Spore	Schaeffer-Fulton — malachite green (heat) + safranin → green spore, red vegetative cell
Flagella	Leifson / Ryu — mordant-thickened; Fontana for spirochaetes (silver impregnation)
Metachromatic granules	Albert's, Neisser's, Loeffler's methylene blue — <i>Corynebacterium</i> (Babes-Ernst granules)
Giemsa	Blood parasites, <i>Anaplasma</i> , <i>Chlamydia</i> inclusions, <i>Haemobartonella</i>

Microscopy & micrometry

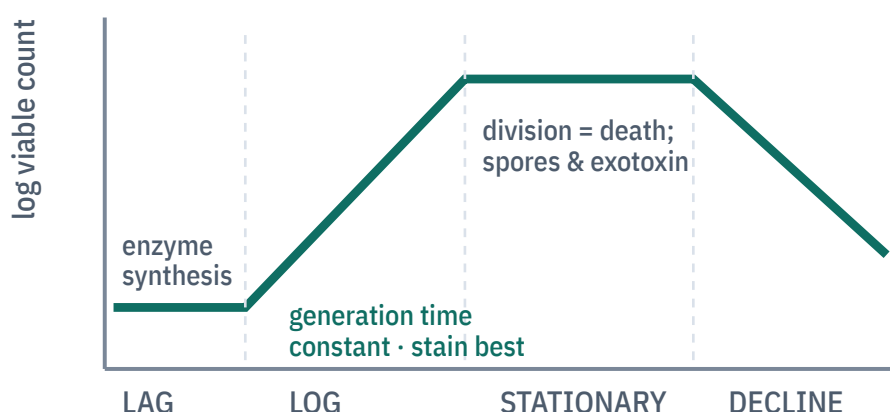
RAPID-FIRE

Resolving power	$RP = 0.61\lambda / NA$
Light micro. limit	$\approx 0.2 \mu\text{m}$; useful mag $\approx 1000\text{--}1500\times$
Oil immersion	Cedar wood oil, RI 1.515 = glass; raises NA to 1.25–1.4
Dark field	Live, unstained — <i>Leptospira</i> , <i>Treponema</i>
Phase contrast	Live unstained cells, internal detail
Fluorescence	Auramine-rhodamine for AFB; FAT for rabies
Electron micro.	TEM resolves $\approx 0.2 \text{ nm}$ — only way to see virions
Micrometry	Stage micrometer calibrates the ocular micrometer; 1 stage div = $10 \mu\text{m}$

Calibration 1 ocular div (μm) = (stage divs coinciding $\times 10$) $\div$ ocular divs coinciding. Recalibrate for *every* objective.

Growth curve & requirements

MECHANISM



- **Generation time:** *E. coli* 20 min · *M. tuberculosis* 18–24 h · *M. leprae* ~13 d.
- Temperature: psychrophile <20 · **mesophile 25–40 (all pathogens)** · thermophile 55–80.
- pH: most 7.2–7.4; *Vibrio* alkaline 8.5–9.0; fungi acidic 5.6.
- O₂: obligate aerobe · facultative anaerobe · **microaerophile (*Campylobacter* 5% O₂)** · obligate anaerobe · capnophile (5–10% CO₂ — *Brucella abortus*).

Media — the exam list

RAPID-FIRE

Basal	Nutrient broth/agar, peptone water
Enriched	Blood agar, chocolate agar, serum agar
Enrichment (liquid)	Selenite F, tetrathionate, alkaline peptone water
Selective	MacConkey, DCA, TCBS, Lowenstein-Jensen, SDA
Differential	MacConkey (lactose), blood agar (haemolysis), EMB

Indicator	Wilson-Blair (black colonies of <i>Salmonella</i>)
Transport	Cary-Blair, Stuart's, Amies
Anaerobic	Robertson's cooked meat, thioglycollate, Brewer's
Sugar-free	For toxin studies of <i>C. diphtheriae</i>

Anaerobiosis McIntosh–Fildes jar · GasPak · Robertson's meat (unsaturated fat + glutathione) · Methylene blue indicator: blue = O₂ present, colourless = anaerobic.

Normal, opportunistic & saprophytic flora

TABLE

Site	Predominant normal flora	Note
Skin	<i>Staphylococcus</i> (coag -ve), <i>Micrococcus</i> , <i>Corynebacterium</i>	Source of surgical-site infection
Rumen	Cellulolytic bacteria, protozoa, anaerobic fungi	Symbiotic – VFA supply
Intestine	<i>E. coli</i> , <i>Lactobacillus</i> , <i>Bacteroides</i> , <i>Clostridium</i>	Vitamin K & B synthesis; colonisation resistance
Upper resp.	<i>Pasteurella</i> , <i>Mannheimia</i> , streptococci	Stress → shipping fever
Teat canal	CNS, <i>Corynebacterium bovis</i>	Keratin plug is first defence
Sterile sites	Blood, CSF, muscle, internal organs, urinary bladder, uterus of healthy animal	

Define cleanly: **Commensal** = benefits, host unharmed. **Opportunist** = normal flora turning pathogen when host defence falls (immunosuppression, catheter, prolonged antibiotics). **Saprophyte** = lives on dead organic matter, rarely invades.

History – dates that get asked

RAPID-FIRE

Leeuwenhoek 1676	First saw "animalcules"	Jenner 1796	Cowpox → smallpox vaccine
Pasteur	Germ theory, attenuation, anthrax & rabies vaccine, pasteurisation	Koch 1876/1882	<i>B. anthracis</i> ; tubercle bacillus; solid media, postulates
Lister 1867	Antisepsis – carbolic acid	Ehrlich	Salvarsan; side-chain theory
Fleming 1928	Penicillin from <i>P. notatum</i>	Iwanowski 1892	TMV – birth of virology
Gram 1884	Differential staining	Petri 1887	Petri dish

Koch's postulates – write all four

1. Organism present in *every* case, absent from healthy animals.
2. Isolated and grown in **pure culture**.
3. Pure culture reproduces the same disease in a susceptible host.
4. Re-isolated from the experimentally infected host.

Exceptions to quote: *M. leprae* & *T. pallidum* (not cultivable), viruses/prions, carriers, polymicrobial & toxin-mediated disease. Modern answer: **molecular Koch's postulates** (gene knockout abolishes virulence, restoration returns it).

Classification & nomenclature

RAPID-FIRE

Reference	Bergey's Manual of Systematic Bacteriology – the standard
Binomial rule	Genus capitalised, species lower case, both italic/underlined: <i>Staphylococcus aureus</i> ; abbreviate <i>S. aureus</i> after first use
Hierarchy	Domain→Phylum→Class→Order→Family→Genus→Species
Species (bacteria)	Strains with $\geq 70\%$ DNA–DNA hybridisation, $< 5^\circ\text{C } \Delta T_m$
Gold standard now	16S rRNA gene sequencing (~1500 bp; $> 97\%$ identity = same species)
Sub-species terms	Serovar (antigen) · biovar (biochemical) · pathovar · phagovar · biotype
Type strain	Nomenclatural reference; deposited (ATCC, NCTC, MTCC)
Other schemes	Numerical taxonomy (Adansonian), chemotaxonomy, MLST, MALDI-TOF MS

Sheet 01 • 60-second self-test

RECALL

Prompt	Answer
Mordant of Gram stain	Gram's iodine
Chemical unique to bacterial spore	Calcium dipicolinate
Toxic moiety of endotoxin	Lipid A
Stain for anthrax capsule	McFadyean's
Acid used in modified ZN	0.5–1% H_2SO_4
Enzyme cross-linking peptidoglycan	Transpeptidase (PBP)
Phase with maximum exotoxin/spores	Stationary phase
Ribosome of bacteria	70S = 50S + 30S
Medium for <i>Vibrio</i>	TCBS / alkaline peptone water
Anaerobic indicator dye	Methylene blue

Infection – the vocabulary that carries marks

RAPID-FIRE

Infection	Entry, multiplication & establishment of an agent in host	Contamination	Presence on surface, no multiplication
Primary	First infection in a fresh host	Secondary	New agent on a weakened host
Focal	Confined to one site	Cross	New agent acquired from another patient/hospital
Nosocomial	Hospital-acquired	Iatrogenic	Caused by a procedure/physician
Latent	Agent persists silently, reactivates	Sub-clinical	Infected, no signs, still sheds
Zoonosis	Animal → human (anthrax, brucellosis, TB, leptospirosis)	Carrier	Healthy, incubatory, convalescent, chronic, contact

Route	How it happens	Example
Ingestion	Contaminated feed/water/milk	Salmonellosis, Johne's
Inhalation	Aerosol, droplet nuclei	Tuberculosis, pasteurellosis
Contact	Direct / fomite / venereal	Dermatophilosis, campylobacteriosis
Vector	Biological or mechanical arthropod	Anaplasmosis, HS spread by flies
Vertical	Transplacental, transovarian, milk	Brucellosis, <i>S. pullorum</i>
Iatrogenic	Needles, instruments, surgery	Tetanus, abscess

Pathogenicity → disease

MECHANISM

1 • ADHESION & COLONISATION

Fimbriae (K88/F4, K99/F5, 987P), LTA, biofilm



2 • INVASION

Invasins, T3SS injectisome, hyaluronidase, collagenase



3 • EVASION OF DEFENCE

Capsule (antiphagocytic), protein A, coagulase, IgA protease, catalase, antigenic variation, intracellular survival



4 • IRON ACQUISITION

Siderophores — enterobactin, aerobactin; Tf/Lf receptors



5 • DAMAGE

Exotoxin, endotoxin, enzymes, immunopathology



DISEASE

Define Pathogenicity = *qualitative* ability of a species to cause disease. **Virulence** = *quantitative* degree of pathogenicity, measured as LD_{50} / ID_{50} . Lower LD_{50} = more virulent.

Exotoxin vs Endotoxin

TABLE

Feature	Exotoxin	Endotoxin
Nature	Protein (polypeptide)	LPS — Lipid A
Source	Mostly Gram +ve, some -ve; secreted	Gram -ve outer membrane; released on lysis
Heat	Labile (60 °C/30 min) — except staph enterotoxin	Stable (121 °C, 1 h)
Potency	Highly potent (botulinum most potent known)	Weak, needs large dose
Specificity	Specific receptor & target tissue	Non-specific — fever, DIC, shock
Antigenicity	Strong; neutralised by antitoxin	Poor; antibody not protective
Toxoid	Yes (formalin 0.3–0.4%, 37 °C)	Cannot be toxoided
Genes	Plasmid / lysogenic phage often	Chromosomal

Feature	Exotoxin	Endotoxin
Detection	Neutralisation, ELISA, mouse test	LAL (<i>Limulus amoebocyte lysate</i>)

Also define: **Bacteraemia** = organisms in blood, transient, no multiplication. **Septicaemia** = multiplication in blood + systemic signs. **Toxaemia** = toxin in blood without the organism (tetanus, enterotoxaemia). **Pyaeemia** = pus-forming organisms in blood → metastatic abscesses.

Named toxins to memorise

RAPID-FIRE

A-B model	B = binding subunit, A = active enzymatic subunit
Diphtheria & <i>Pseudomonas</i> ExoA	ADP-ribosylate EF-2 → protein synthesis stops
Cholera / <i>E. coli</i> LT	ADP-ribosylate Gs → ↑ cAMP → secretory diarrhoea
<i>E. coli</i> ST	↑ cGMP via guanylate cyclase C
Tetanospasmin	Cleaves synaptobrevin → blocks glycine/GABA → spastic paralysis
Botulinum	Cleaves SNARE → blocks ACh release → flaccid paralysis
Anthrax	PA + EF (adenylate cyclase, oedema) + LF (protease, MAPKK)
α-toxin <i>C. perfringens</i>	Lecithinase (phospholipase C) — Nagler reaction
ε-toxin <i>C. perfringens</i> D	Pulpy kidney; permease, ↑ vascular permeability
Staph α / PVL / TSST-1	Pore-forming / leucocidin / superantigen
Streptolysin O & S	O = oxygen-labile, antigenic (ASO titre); S = stable, non-antigenic
Superantigen	Bridges MHC-II to TCR Vβ outside groove → mass T-cell activation, cytokine storm

Gene transfer — the three routes

MECHANISM

TRANSFORMATION

Naked DNA taken up by *competent* cell · Griffith 1928
Avery, MacLeod & McCarty 1944

TRANSDUCTION

Bacteriophage carries DNA
Generalised — lytic, any gene
Specialised — lysogenic, adjacent genes
Zinder & Lederberg 1952

CONJUGATION

Sex pilus, cell contact · $F^+ \rightarrow F^-$
Hfr = integrated F · Lederberg & Tatum 1946

Transposons ("jumping genes") move within/between replicons and carry resistance cassettes.
Integrans capture gene cassettes by site-specific recombination — the engine of multi-drug R plasmids.

Mutation & plasmids

RAPID-FIRE

Point mutation	Substitution — missense, nonsense, silent
Frameshift	Insertion / deletion (not ×3)

Spontaneous rate	$10^{-6} - 10^{-9}$ per cell per division
Mutagens	UV (thymine dimers), alkylating agents, base analogues, acridine dyes
Ames test	Mutagenicity screen using <i>S. typhimurium</i> his ⁻
F plasmid	Fertility — conjugation
R plasmid	RTF + r-determinant; multidrug resistance
Col plasmid	Bacteriocins (colicin)
Virulence plasmid	pXO1/pXO2 anthrax; ETEC enterotoxin; <i>Yersinia</i> pYV
Episome	Plasmid able to integrate into chromosome
Curing	Loss of plasmid — acridine orange, ethidium bromide, heat

Antibiotic resistance

MECHANISM

Mechanism	Example
Enzymatic destruction	β -lactamase, ESBL, AmpC , carbapenemase (NDM-1); acetyl/adenyl transferase for aminoglycosides
Target alteration	PBP2a (<i>mecA</i>) = MRSA ; ribosomal methylation (<i>erm</i>); DNA gyrase mutation → quinolone R
Reduced uptake	Porin loss (<i>Pseudomonas</i>)
Efflux pumps	Tetracycline (<i>tet</i>), multidrug RND pumps
Bypass pathway	Sulphonamide resistance via exogenous folate
Biofilm / persisters	Phenotypic tolerance, not genetic

Vet relevance Sub-therapeutic in-feed antibiotics & poor withdrawal-period compliance drive AMR into the food chain. **MRSA, ESBL *E. coli*, MDR *Salmonella*** are the named "One Health" examples. India: NAP-AMR; ban on colistin as growth promoter (2019).

Long-answer skeleton – "Virulence determinants of bacteria"

SKELETON

Ten-line frame that fits any 10–15 mark question

1. **Define** pathogenicity vs virulence; state LD₅₀/ID₅₀.
2. **Adhesins** — fimbrial (K88, K99, 987P, F41) and afimbrial; tissue tropism.
3. **Invasion factors** — invasins, T3SS, spreading enzymes (hyaluronidase = "spreading factor", collagenase, fibrinolysin).
4. **Antiphagocytic** — capsule, M protein, protein A, coagulase, leucocidin.
5. **Intracellular survival** — inhibition of phagolysosome fusion (*M. tuberculosis*), escape to cytosol (*Listeria*, listeriolysin O), catalase/SOD.
6. **Iron piracy** — siderophores.
7. **Toxins** — exotoxin classes (neuro-, entero-, cyto-, superantigen) + endotoxin.
8. **Antigenic variation & molecular mimicry**.
9. **Biofilm** and quorum sensing.
10. **Genetic basis** — pathogenicity islands, plasmids, lysogenic conversion; close with one Indian field example.

Sheet 02 · 60-second self-test

RECALL

Prompt	Answer
Most potent bacterial toxin	Botulinum
Toxin causing spastic paralysis	Tetanospasmin
Test to detect endotoxin	LAL test
Gene for methicillin resistance	<i>mecA</i> → PBP2a
Transduction discovered by	Zinder & Lederberg, 1952
Transformation discovered by	Griffith, 1928
Plasmid-free strain produced by	Curing (acridine orange)
Enzyme called "spreading factor"	Hyaluronidase
Toxin ADP-ribosylating EF-2	Diphtheria toxin
Iron-chelating virulence molecule	Siderophore

Sorting the Gram-positives at the bench

MECHANISM

Gram-positive isolate

COCCI → test: CATALASE

Catalase +ve

Clusters · *Staphylococcus*
 Coagulase + → *S. aureus*
 Coagulase – → CNS (*S. epidermidis*)

Catalase –ve

Chains · *Streptococcus*
 β -haem → *S. agalactiae*, *S. equi*
 Bile-esculin + → *Enterococcus*

RODS → test: SPORES?

Spore-forming

Aerobe → *Bacillus*
 Anaerobe → *Clostridium*
 (see sheet 04)

Non-sporing

Pleomorphic, granules → *Coryne*.
 Tumbling motility → *Listeria*
 H_2S in TSI, non-motile → *Erysipelothrix*

Staphylococcus

SYSTEMATIC

- **Morphology:** Gram +ve cocci ~1 μm in **grape-like clusters**; non-motile, non-sporing, catalase +ve, facultative anaerobe.
- **Culture:** Golden-yellow colonies on nutrient agar; **β -haemolysis** on blood agar; grows in **7.5–10% NaCl** (Mannitol Salt Agar – mannitol fermented, medium turns yellow); Baird-Parker = black shiny colonies with clear halo.
- **Key test: Coagulase** — slide test detects bound clumping factor, tube test detects free coagulase. Coagulase +ve = pathogenic *S. aureus*.
- **Other tests:** DNase +, phosphatase +, protein A present, ferments mannitol anaerobically.
- **Virulence:** Protein A (binds Fc of IgG), coagulase, hyaluronidase, staphylokinase, lipase, $\alpha/\beta/\gamma/\delta$ haemolysins, **Panton-Valentine leucocidin**, exfoliative toxin, **TSST-1 & enterotoxins A–E (superantigens, heat-stable)**.
- **Veterinary diseases:** bovine mastitis (chief contagious cause, hard fibrotic quarter, gangrenous mastitis in sheep = "blue bag"), tick pyaemia in lambs, **bumblefoot** and synovitis in poultry, botryomycosis, pyoderma (*S. pseudintermedius* in dogs), exudative epidermitis/greasy pig disease (*S. hyicus*).
- **Public health:** staphylococcal food poisoning — 2–6 h, vomiting, from enterotoxin in milk products; **MRSA** (*mecA*).

Differentiate: *Staphylococcus* catalase +, clusters, coagulase $\pm$ · *Micrococcus* catalase +, oxidative, modified oxidase + · *Streptococcus* catalase –, chains.

Streptococcus & Enterococcus

SYSTEMATIC

- **Morphology:** Gram +ve cocci in **chains/pairs**, non-motile, **catalase –ve**, capnophilic, need enriched media.
- **Haemolysis classification:** α = partial, green (*S. pneumoniae*, viridans) · β = complete, clear (*S. agalactiae*, *S. equi*) · γ = none (*Enterococcus*).
- **Lancefield grouping** — precipitation against group-specific **C carbohydrate** of the cell wall (Rebecca Lancefield).

Group	Species	Disease
A	<i>S. pyogenes</i>	Human; strangles-like in primates
B	<i>S. agalactiae</i>	Contagious bovine mastitis — lives in udder, spread at milking
C	<i>S. equi</i> subsp. <i>equi</i>	Strangles — abscessed submandibular & retropharyngeal nodes
C	<i>S. dysgalactiae</i>	Environmental/summer mastitis, polyarthritis in lambs
D	<i>Enterococcus faecalis/faecium</i>	Enteric, UTI, VRE; bile-esculin +, 6.5% NaCl +
E	<i>S. porcinus</i>	Jowl abscess in pigs
—	<i>S. uberis</i> , <i>S. suis</i>	Environmental mastitis; meningitis in piglets (zoonotic)

Tests CAMP test + = *S. agalactiae* (arrowhead haemolysis with *S. aureus*) · Bacitracin sensitive = group A · Optochin sensitive & bile soluble = *S. pneumoniae* · **Hippurate hydrolysis + = *S. agalactiae*.**

Corynebacterium , Trueperella , Rhodococcus

SYSTEMATIC

- **Corynebacterium** — Gram +ve, **pleomorphic club-shaped** rods in **Chinese-letter / palisade / V-L** arrangement; **metachromatic (Babes-Ernst) granules** shown by Albert's / Neisser's stain; non-motile, non-sporing, catalase +.
- *C. pseudotuberculosis* → **caseous lymphadenitis** of sheep & goats (onion-ring greenish caseous nodes), ulcerative lymphangitis in horses. Toxin: **phospholipase D**. Reverse CAMP with *R. equi*.
- *C. renale* group → **bovine pyelonephritis & cystitis**; pizzle rot (ulcerative posthitis) in wethers; urease strongly +.
- *C. diphtheriae* → diphtheria in man; toxin is phage(β)-coded; Loeffler's serum slope, potassium tellurite (black colonies); **Elek's gel precipitation test** for toxigenicity.
- **Trueperella pyogenes** (ex-Arcanobacterium/*Corynebacterium pyogenes*) — the classic **pus-forming** organism of cattle: summer mastitis, abscess, endometritis, pneumonia, arthritis. Small β -haemolytic colonies at 48 h, CAMP +.
- **Rhodococcus equi** — **partially acid-fast** coccobacillus, salmon-pink mucoid colonies, urease +. **Suppurative bronchopneumonia with abscesses in foals 1–4 months**; ulcerative enteritis. Virulence plasmid **VapA**; survives inside macrophages.

Listeria vs Erysipelothrix

TABLE

Feature	<i>L. monocytogenes</i>	<i>E. rhusiopathiae</i>
Motility	Tumbling at 25 °C, umbrella growth in semisolid	Non-motile
Catalase	Positive	Negative
H ₂ S in TSI	Negative	Positive
Haemolysis	Narrow β (listeriolysin O); CAMP +	α on blood agar
Growth trick	Cold enrichment 4 °C — psychrotroph	"Test-tube brush" in gelatin stab
Disease	Circling disease/encephalitis (ruminants), abortion, septicaemia in monogastrics; silage disease	Swine erysipelas — rhomboid "diamond skin"; arthritis, endocarditis; ovine polyarthritis; turkey septicaemia
Zoonosis	Listeriosis — abortion, meningitis (immunocompromised)	Erysipeloid — fish-handler's / whale-finger
Immunity	Classic model of cell-mediated immunity ; intracellular, ActA-driven actin tails	Neuraminidase, hyaluronidase; bacterin/live vaccine

Trap: *Listeria* is a facultative **intracellular** pathogen — escapes phagosome using **listeriolysin O** and moves cell to cell with actin tails, so antibody is nearly useless and **T-cell immunity** is the answer.

Mastitis pathogens — the table examiners love

TABLE

Type	Organism	Reservoir & spread	Signature
Contagious	<i>Staph. aureus</i>	Infected udder → milking	Chronic, fibrosis, gangrene
	<i>Strep. agalactiae</i>	Obligate udder parasite	High SCC, eradicable
	<i>Mycoplasma bovis</i>	Udder, respiratory	Multiple quarters, no fever response to antibiotics
Environmental	<i>E. coli</i> , <i>Klebsiella</i>	Bedding, faeces	Acute toxic (endotoxic) mastitis
	<i>Strep. uberis/dysgalactiae</i>	Environment, skin	Clinical, dry-period onset
	<i>T. pyogenes</i> + <i>F. necrophorum</i>	Flies (<i>Hydrotaea</i>)	Summer mastitis — foul-smelling pus, quarter lost

Screening CMT / SFMT (California & Surf Field Mastitis Test) — detects DNA of somatic cells; **SCC > 2 lakh/ml** = subclinical mastitis. Confirm by culture + ABST before dry-cow therapy.

Sheet 03 • 60-second self-test

RECALL

Prompt	Answer
Single best test to separate Staph from Strep	Catalase

Prompt	Answer
Marker of pathogenic <i>S. aureus</i>	Coagulase
Selective medium for staphylococci	Mannitol salt / Baird-Parker
Lancefield antigen	Cell wall C carbohydrate
CAMP test positive species	<i>S. agalactiae</i> , <i>L. monocytogenes</i>
Cause of strangles	<i>S. equi</i> subsp. <i>equi</i>
Caseous lymphadenitis toxin	Phospholipase D
Foal pneumonia with abscesses	<i>Rhodococcus equi</i> (VapA)
Cold enrichment organism	<i>Listeria monocytogenes</i>
Diamond skin disease	<i>Erysipelothrix rhusiopathiae</i>

04 Spore-Formers, Acid-Fast & Filamentous Bacteria

t17–t20

Bacillus anthracis – the complete answer

SYSTEMATIC

- **Morphology:** large Gram +ve rod 1–1.5 × 3–8 µm with **square/truncate ends**; in culture long chains = **bamboo-stick / boxcar**; capsule of **poly-D-glutamic acid** (pXO2); **central, ellipsoidal, non-bulging spore**.
- **Golden rule:** spores form **only outside the body in free oxygen** — never in a living host or unopened carcass. **Never necropsy a suspect case**; spores survive in soil 50+ years ("incubator areas").
- **Culture:** non-haemolytic on sheep blood agar (*B. cereus* is β-haemolytic); **Medusa-head** colony edge; **inverted fir-tree** in gelatin stab; **string-of-pearls** with 0.5 IU/ml penicillin; grows on bicarbonate agar + 5% CO₂ to make capsule.
- **Toxin (pXO1, tripartite, A-B type):** **PA** (binds CMG2/TEM8, forms heptamer pore) + **EF** (calmodulin-dependent adenylate cyclase → oedema) + **LF** (Zn-metalloprotease cleaving MAPKK → macrophage death, shock).
- **Virulence needs BOTH plasmids.** Sterne 34F₂ vaccine = pXO2[−] (non-capsulated, toxigenic, live spore) — the vaccine used in India. Pasteur's was heat-attenuated.
- **Diagnosis:** peripheral blood/ear-vein smear with **McFadyean's polychrome methylene blue** — blue rods with pink capsule; **Ascoli thermoprecipitation test** on decomposed tissue/hides; PCR for *pagA*, *capB*; culture; animal inoculation in guinea pig.
- **Clinical:** peracute sudden death in cattle/buffalo/sheep; **dark tarry unclotted blood from natural orifices, no rigor mortis, enlarged friable "blackberry-jam" spleen**. Man: cutaneous **malignant pustule**, woolsorter's disease, intestinal.
- **Control:** do not open carcass; burn or bury 6 ft deep with quicklime; 10% formaldehyde/5% NaOH; annual Sterne vaccination in endemic belt; notifiable.

***B. anthracis* vs *B. cereus*:** non-motile / motile · non-haemolytic / β-haemolytic · capsulated / non-capsulated · penicillin-sensitive / resistant · susceptible to gamma phage & string-of-pearls / not. *B. cereus* = food poisoning (emetic & diarrhoeal) and bovine mastitis.

Mycobacterium

SYSTEMATIC

- **Acid-fastness** is due to **mycolic acid** in a lipid-rich (up to 60%) wall + peptidoglycolipid (**wax D**), demonstrated by **Ziehl-Neelsen**. Slender, beaded, non-motile, non-sporing, obligate aerobe.
- **Media: Lowenstein-Jensen** (egg, malachite green, glycerol) · Stonebrink (pyruvate) for *M. bovis* · Middlebrook 7H10/7H11 · **Dorset egg**, Herrold's egg-yolk with mycobactin J for *M. paratuberculosis*.
- **Growth:** slow — *M. tuberculosis* 2–4 wks (eugonic, rough buff tough), *M. bovis* 6–8 wks (**dysgonic**, needs pyruvate, glycerol inhibits), *M. avium* 2–3 wks, *M. paratuberculosis* 8–16 wks (**mycobactin-dependent** — the defining test).
- **Runyon groups** (atypical/NTM): I photochromogen · II scotochromogen · III non-chromogen · IV rapid grower (*M. fortuitum*).
- **Bovine TB (*M. bovis*):** tubercle = caseation + calcification; miliary or organ TB; **zoonotic via raw milk**. Diagnosis: **single intradermal / single intradermal comparative cervical tuberculin test** (avian vs bovine PPD, read at 72 h), **IFN-γ (Bovigam) assay**, ZN of milk/tissue, culture, PCR. Test-and-slaughter policy.
- **Johne's disease (*M. avium* subsp. *paratuberculosis*):** chronic granulomatous enteritis of cattle & buffalo; incubation 2–5 yrs; **corrugated, thickened "brain-like" ileum**, no ulcer; intermittent pipe-stream diarrhoea, emaciation, submandibular oedema, normal appetite. Diagnosis: ZN of rectal pinch/faeces (clumps of AFB), **johnin test**, AGID, ELISA, faecal culture, IS900 PCR.
- **Avian TB (*M. avium*):** nodules in liver, spleen, intestine **without calcification**; positive avian tuberculin.
- **Virulence:** cord factor (trehalose-6,6'-dimycolate), sulpholipids, LAM, catalase-peroxidase; blocks phagosome-lysosome fusion → survives in macrophage → **Type IV hypersensitivity & granuloma**.

Clostridium — split by what the toxin does

TABLE

Group	Species	Disease & key toxin
Neurotoxic	<i>C. tetani</i>	Tetanus — drumstick/tennis-racquet terminal round spore, swarming, tetanospasmin blocks glycine/GABA at Renshaw cells → spastic paralysis, lockjaw, sawhorse stance, third-eyelid prolapse. Toxoid (ATT) + antitoxin
	<i>C. botulinum</i>	Botulism — types A–G; C & D in animals; preformed toxin in spoiled silage/carcass; flaccid paralysis, "limberneck" in birds. Type C = "lamziekte" with osteophagia in P-deficient cattle
Histotoxic	<i>C. chauvoei</i>	Black quarter/blackleg — endogenous, crepitant emphysematous myositis of heavy muscle in young cattle; rancid butter smell
	<i>C. septicum</i> / <i>novyi</i> / <i>sordellii</i>	Malignant oedema, braxy (abomasitis), black disease (<i>C. novyi</i> B with <i>Fasciola</i> migration), big head of rams, gas gangrene

Group	Species	Disease & key toxin
Enterotoxigenic	<i>C. perfringens</i> A–E	A gas gangrene, yellow lamb disease · B lamb dysentery · C struck, necrotic enteritis · D pulpy kidney/enterotoxaemia (ϵ-toxin) , glycosuria · E enteritis. Type A also = necrotic enteritis in broilers
	<i>C. difficile</i>	Toxins A & B; colitis in horses/pigs after antibiotics

Toxin typing α =lecithinase (all types, **Nagler reaction** on egg-yolk agar, half-plate antitoxin) · β =B,C (trypsin-labile — why lambs on colostrum are hit) · ϵ =B,D · ι =E. Confirm by mouse neutralisation / multiplex PCR.

Blackleg vs malignant oedema: BQ = endogenous, no wound, heavy muscles, *C. chauvoei*. MO = wound-associated, subcutaneous, *C. septicum* chiefly, oedema > gas.

Actinomyces , Nocardia , Streptomyces , Dermatophilus

TABLE

Genus	O ₂ & acid-fast	Disease
<i>Actinomyces bovis</i>	Anaerobe/capnophile · Not acid-fast	Lumpy jaw — pyogranulomatous osteomyelitis of mandible/maxilla in cattle; sulphur granules in pus (club colonies)
<i>Nocardia asteroides</i>	Strict aerobe · Partially acid-fast (modified ZN)	Bovine nocardial mastitis, canine/feline pyothorax, granulomatous pneumonia; branching filaments, soil saprophyte
<i>Streptomyces</i>	Aerobe · not acid-fast	Mostly soil saprophyte; source of most antibiotics (streptomycin, tetracycline); rare mycetoma
<i>Dermatophilus congolensis</i>	Facultative anaerobe · not acid-fast	Dermatophilosis / rain scald / strawberry footrot ; branching filaments dividing in two planes → " railroad-track " parallel rows of cocci (zoospores) ; motile flagellated zoospores activated by moisture

Actinobacillosis vs Actinomycosis: *Actinobacillus lignieresii* = **wooden tongue**, Gram –ve, soft tissue. *Actinomyces bovis* = **lumpy jaw**, Gram +ve, bone. Both give sulphur granules — the Gram reaction of the granule is the differentiator.

Sterilisation & disinfection — one-glance table

TABLE

Method	Condition	Use
Hot air oven	160 °C / 2 h	Glass, oil, powder
Autoclave	121 °C, 15 psi, 15 min	Media, instruments; gold standard
Tyndallisation	100 °C / 20 min × 3 d	Sugars, gelatin
Inspissation	80–85 °C / 30 min × 3 d	LJ, Loeffler serum
Pasteurisation	LTLT 63 °C/30 min · HTST 72 °C/15 s	Milk — kills <i>M. bovis</i> , <i>Coxiella</i>

Method	Condition	Use
Filtration	0.22 µm membrane	Sera, antibiotics (not viruses)
Radiation	UV 260 nm (surface) · gamma (ionising)	Cabinets · plasticware
Chemical	Ethylene oxide, glutaraldehyde, formalin, phenol, hypochlorite, quaternary ammonium	Heat-labile items, premises

Controls Autoclave → *Geobacillus stearothermophilus* spores · Hot air oven → *Bacillus atrophaeus* (*subtilis*) spores · Ethylene oxide → *B. atrophaeus*. Browne's tube & autoclave tape = chemical indicators. **Rideal-Walker / Chick-Martin** = phenol coefficient.

Sheet 04 · 60-second self-test

RECALL

Prompt	Answer
Anthrax capsule chemistry	Poly-D-glutamic acid (pXO2)
Anthrax toxin components	PA + EF + LF
Anthrax field test on hides	Ascoli thermoprecipitation
Anthrax vaccine strain	Sterne 34F ₂ (pXO2 ⁻)
Chemical basis of acid-fastness	Mycolic acid
Growth factor for <i>M. paratuberculosis</i>	Mycobactin J
PCR target for Johne's	IS900
Terminal round spore, drumstick	<i>C. tetani</i>
Nagler reaction detects	<i>C. perfringens</i> α-toxin (lecithinase)
Pulpy kidney toxin	ε-toxin, <i>C. perfringens</i> type D
Sulphur granules, Gram +ve, bone	<i>Actinomyces bovis</i>
Railroad-track zoospores	<i>Dermatophilus congolensis</i>

Identifying a Gram-negative rod in six moves

MECHANISM

Gram -ve rod on MacConkey → test: OXIDASE

OXIDASE -ve • Enterobacteriaceae •
glucose fermenter • nitrate → nitrite →
test: LACTOSE

Lactose FERMENTER

Pink on MacConkey

E. coli (IMViC ++-- , green sheen EMB)*Klebsiella* (mucoid, non-motile, VP+)**NON-fermenter**

Pale colonies

H₂S+, motile → *Salmonella*Swarms, urease++ → *Proteus*; *Yersinia* 22 °C
motile

OXIDASE +ve • non-enteric → test: GROWTH
ON MacCONKEY

GrowsPigment, grape odour → *Pseudomonas*Safety-pin, glanders → *Burkholderia*Curved, TCBS → *Vibrio***Poor / no growth**Bipolar coccobacillus → *Pasteurella*X & V factors → *Haemophilus*CO₂-dependent → *Brucella*; seagull →
*Campylobacter**E. coli* , *Klebsiella* & the pathotypes

SYSTEMATIC

- **Family traits:** Gram -ve rods, **oxidase -ve**, ferment glucose, reduce nitrate, catalase +, facultative anaerobe. Antigens **O (somatic, LPS, heat-stable)** • **H (flagellar, protein, heat-labile)** • **K (capsular)** • **Vi (*S. Typhi*, *Citrobacter*)**.
- ***E. coli*:** motile, lactose +, **IMViC = + + - -**, metallic green sheen on EMB, MR +, VP -, citrate -, indole +.

Pathotype	Mechanism	Veterinary disease
ETEC	Fimbriae K88/F4, K99/F5, 987P, F41 + LT/ST enterotoxin	Neonatal calf/piglet/lamb diarrhoea (secretory, <1 wk)
EPEC	Intimin (<i>eae</i>) → attaching-effacing lesion	Diarrhoea in rabbits, dogs, calves
STEC/VTEC	Shiga toxin Stx1/Stx2	Oedema disease of weaned pigs (Stx2e); O157:H7 HUS in man — cattle are the reservoir
Septicaemic	Capsule, aerobactin, serum resistance	Colisepticaemia in calves (failure of passive transfer), colibacillosis in poultry — pericarditis, perihepatitis, airsacculitis
UPEC / mastitis	P fimbriae; endotoxin	Cystitis, pyometra; acute toxic coliform mastitis

***Klebsiella pneumoniae*: non-motile**, prominent polysaccharide capsule, mucoid stringy colonies, **VP +, citrate +, urease weakly +, indole –** (mirror image of *E. coli*). Environmental mastitis from sawdust bedding, metritis in mares, pneumonia in foals, CAUTI. Multi-drug resistant (ESBL, carbapenemase).

Salmonella , Yersinia , Proteus

SYSTEMATIC

- **Salmonella**: motile (except *S. Gallinarum* & *Pullorum*), non-lactose fermenter, **H₂S +**, urease –, indole –, citrate +. TSI = **alkaline slant / acid butt with blackening**. Media: selenite F / tetrathionate broth → XLD, BSA (Wilson-Blair, **black colonies**), MacConkey pale.
- **Kauffmann-White scheme** classifies >2600 serovars on O, H (phase 1 & 2) and Vi antigens.
- **Diseases**: calf paratyphoid (*S. Dublin*, Typhimurium) — septicaemia, enteritis, abortion; **abortion in mares (*S. Abortusequi*)** and ewes (*S. Abortusovis*); **fowl typhoid (*S. Gallinarum*)** and **pullorum disease/BWD (*S. Pullorum*, egg-transmitted)** — detected by **rapid serum/whole-blood plate agglutination**; swine salmonellosis (*S. Choleraesuis*). Food-borne zoonosis via eggs and meat.
- **Virulence**: SPI-1 & SPI-2 pathogenicity islands (T3SS), Vi capsule, endotoxin, survival in macrophages.
- **Yersinia**: **bipolar staining, motile at 22 °C but non-motile at 37 °C**, urease +. *Y. pestis* — plague, safety-pin appearance, flea-borne, **rodent reservoir**, Fraction 1 capsule, V & W antigens. *Y. enterocolitica* — cold enrichment, enterocolitis, pigs are the reservoir. *Y. pseudotuberculosis* — caseous nodules in rodents, birds, sheep.
- **Proteus**: **swarming** on blood agar (inhibited by 5% agar / CLED), **urease strongly +** (→ struvite urolith & alkaline urine), H₂S +, PDA +, fishy odour. UTI, otitis externa, wound infection. **Weil-Felix** — OX-19/OX-2/OX-K strains cross-react with rickettsial antibodies.

Pasteurella · Mannheimia · Actinobacillus · Haemophilus

TABLE

Organism	Disease & key point
<i>P. multocida</i> B:2 / E:2	Haemorrhagic septicaemia of cattle & buffalo — monsoon, sudden death, hot painful submandibular swelling, petechiae on serosa. Bipolar Gram –ve coccobacillus (Leishman/methylene blue), capsulated, non-motile, indole +, no growth on MacConkey. Vaccine: alum-precipitated / oil-adjuvant HS bacterin
<i>P. multocida</i> A:1	Fowl cholera — greenish diarrhoea, wattle abscess, petechiae in heart fat; also progressive atrophic rhinitis in pigs (with <i>Bordetella</i>) via dermonecrotic toxin
<i>Mannheimia haemolytica</i> A1	Shipping fever / enzootic pneumonia of cattle & sheep; leukotoxin (RTX) kills ruminant leucocytes; β-haemolytic, grows on MacConkey, indole –
<i>Actinobacillus lignieresii</i>	Wooden tongue in cattle — soft-tissue granuloma, sulphur granules, Gram –ve rosettes
<i>A. equuli</i> / <i>A. pleuropneumoniae</i>	Sleepy foal disease (joint-ill, nephritis) · Porcine pleuropneumonia — Apx toxins, NAD-dependent, CAMP +
<i>Haemophilus</i>	Needs X (haemin) and/or V (NAD) factors → satellitism around <i>S. aureus</i> on blood agar. <i>H. parasuis</i> = Glasser's disease (polyserositis, arthritis, meningitis); <i>Avibacterium paragallinarum</i> = infectious coryza in poultry

Brucella

SYSTEMATIC

- Gram -ve, **tiny coccobacilli, non-motile, non-sporing, non-capsulated**, aerobic, **catalase & oxidase +**, **urease +**, **partially acid-fast (modified ZN → red on blue)**.
- **Species & host:** *B. abortus* (cattle) · *B. melitensis* (goat/sheep — most pathogenic for man) · *B. suis* (pig) · *B. ovis* (ram epididymitis) · *B. canis* (dog) · *B. neotomae*, *B. ceti*, *B. pinnipedialis*.
- **Growth aids:** *B. abortus* biovar 1 needs **5–10% CO₂**; differentiate species by CO₂ need, H₂S, urease speed, dye inhibition (**thionine & basic fuchsin**), phage Tbilisi lysis, A/M monospecific sera.
- **Pathogenesis:** facultative intracellular in macrophages & trophoblasts; **erythritol** in the placenta of cattle, sheep, goat, pig favours growth → **abortion in the last trimester**, retained placenta, orchitis/epididymitis, hygroma. Man: **undulant/Malta fever** — a major lab-acquired infection.
- **Tests:** RBPT (screening) · **STAT** · MRT (milk ring, herd screening) · 2-ME/Rivanol (IgG, chronic) · CFT (OIE international trade standard) · **i-ELISA/c-ELISA** · FPA · AGID for *B. ovis* · PCR (*bcs31*, *IS711*) · culture on Brucella selective agar.
- **Vaccine:** *B. abortus* **S19** — live, 4–8 month heifer calves, single dose, gives lifelong immunity but antibody interferes with testing. **RB51** — rough, no smooth-LPS antibody, DIVA-compatible. **Rev-1** for sheep & goats.

Trap: Bull semen and blood are NOT reliable samples — use **aborted foetal stomach content, placental cotyledon, vaginal discharge, milk**. No treatment in animals; policy is test-and-slaughter + vaccination.

Pseudomonas , Vibrio , Campylobacter , Bordetella , Moraxella

RAPID-FIRE

<i>P. aeruginosa</i>	Oxidase +, non-fermenter, pyocyanin (blue) + pyoverdin (green) , sweet grape/corn-tortilla odour, grows at 42 °C. Otitis externa in dogs, mastitis from contaminated water, burn/wound sepsis, notoriously multi-drug resistant ; ExoA toxin
<i>Burkholderia mallei</i>	Glanders in equines — farcy nodules along lymphatics, nasal ulcers; non-motile; Mallein test , CFT; notifiable & zoonotic ; no treatment, destruction policy
<i>B. pseudomallei</i>	Melioidosis — soil/water saprophyte, motile, safety-pin, caseous nodules
<i>Vibrio cholerae</i>	Curved comma rod, single polar flagellum, darting motility , grows at pH 8.5–9.0 , TCBS yellow colonies (sucrose +), cholera toxin ↑ cAMP; O1 (classical, El Tor) & O139
<i>Campylobacter fetus</i>	Seagull-wing / S-shaped , microaerophilic (5% O ₂ , 10% CO ₂), 42 °C for <i>C. jejuni</i> . <i>C. fetus</i> subsp. <i>venerealis</i> = bovine genital campylobacteriosis — infertility, early embryonic death, repeat breeding; subsp. <i>fetus</i> = ovine abortion. <i>C. jejuni</i> = leading food-borne zoonotic enteritis, poultry reservoir, Guillain-Barré sequel
<i>Bordetella bronchiseptica</i>	Motile, oxidase & urease +. Kennel cough complex, atrophic rhinitis in pigs (dermonecrotic toxin + <i>P. multocida</i> toxigenic strain), bronchopneumonia. <i>B. avium</i> = turkey coryza

IMViC & biochemical shorthand

TABLE

Organism	Ind	MR	VP	Cit	Ure	H ₂ S	Mot	Lac	Ox
<i>E. coli</i>	+	+	–	–	–	–	+	+	–
<i>Klebsiella</i>	–	–	+	+	w+	–	–	+	–
<i>Salmonella</i>	–	+	–	+	–	+	+	–	–
<i>Proteus</i>	±	+	–	±	++	+	+	–	–
<i>Yersinia ent.</i>	±	+	–	–	+	–	22°C	–	–
<i>Pseudomonas</i>	–	–	–	+	±	–	+	–	+
<i>P. multocida</i>	+	–	–	–	–	–	–	–	+
<i>M. haemolytica</i>	–	–	–	–	–	–	–	+	+
<i>Brucella</i>	–	–	–	–	+	±	–	–	+

IMViC Indole · Methyl red · Voges-Proskauer · Citrate. *E. coli* = + + – –; *Klebsiella/Enterobacter* = – – + +. That single contrast answers most viva questions on Enterobacteriaceae.

Sheet 05 · 60-second self-test

RECALL

Prompt	Answer
Single test separating enterics from the rest	Oxidase (enterics –ve)
Green metallic sheen on EMB	<i>E. coli</i>
Fimbria of calf-scour ETEC	K99 / F5
Oedema disease toxin	Stx2e (STEC)
Salmonella serotyping scheme	Kauffmann-White
Non-motile salmonellae	<i>S. Gallinarum</i> & <i>Pullorum</i>
Swarming, urease ++	<i>Proteus</i>
Weil-Felix strains	OX-19, OX-2, OX-K of <i>Proteus</i>
Placental growth factor for <i>Brucella</i>	Erythritol
DIVA brucella vaccine	RB51
Leukotoxin of shipping fever	<i>Mannheimia haemolytica</i> RTX
Mallein test diagnoses	Glanders (<i>B. mallei</i>)

Gram-negative anaerobes

SYSTEMATIC

Organism	Veterinary significance
<i>Fusobacterium necrophorum</i>	Long filamentous Gram –ve rod. Calf diphtheria (necrotic laryngitis), hepatic abscess of feedlot cattle (with <i>T. pyogenes</i>), necrotic rumenitis after grain overload, foot rot / interdigital necrobacillosis , summer mastitis, thrush. Leukotoxin + endotoxin
<i>Dichelobacter nodosus</i>	Large Gram –ve rod with swollen ends. Essential (transmitting) agent of ovine foot rot ; <i>F. necrophorum</i> is the predisposing invader. Produces keratinase/protease that digests hoof horn. Vaccine = pilus-based; footbath 10% ZnSO ₄ / 5% formalin
<i>Bacteroides fragilis</i> group	Bile-resistant, non-sporing anaerobe; normal gut flora turned opportunist – abdominal abscess, peritonitis, aspiration pneumonia. Intrinsically β -lactamase producing; metronidazole is the drug
<i>Porphyromonas/Prevotella</i>	Black-pigmented anaerobes; periodontal disease of dog & cat

Anaerobic culture Freshly prepared, pre-reduced media · Robertson's cooked meat · thioglycollate · McIntosh-Fildes jar or GasPak (H₂ + CO₂, palladium catalyst) · anaerobic glove box. Foul putrid odour, gas, growth **away** from the surface in a shake tube.

Leptospira & other spirochaetes

SYSTEMATIC

- **Leptospira interrogans** (pathogenic) vs *L. biflexa* (saprophytic). Thin, tightly coiled, **hooked ends** (question-mark shape), motile by two axial filaments/endoflagella; **too thin for Gram stain** – use **dark-field microscopy**, silver impregnation (Levaditi/Fontana), FAT.
- **Culture: EMJH** or Korthof's/Fletcher's semisolid at 28–30 °C, incubate up to **6–8 weeks**; growth as a Dinger's ring.
- **Serovars & hosts:** Pomona & Hardjo (cattle, pigs) · Canicola & Icterohaemorrhagiae (dog, rodent reservoir) · Grippotyphosa · Bratislava (horses). **Rodents are maintenance hosts and shed lifelong in urine.**
- **Disease:** leptospiraemia → fever, haemoglobinuria ("red water"), jaundice, **abortion & agalactia (milk drop syndrome, flabby-bag)**, interstitial nephritis, recurrent uveitis (periodic ophthalmia) in horses. Man: Weil's disease – occupational zoonosis of paddy workers, sewer workers, veterinarians.
- **Diagnosis: MAT** (microscopic agglutination test – reference standard, titre $\geq 1:100$ or 4-fold rise), dark-field of fresh urine, ELISA, PCR (*lipL32*, 16S). Treat with streptomycin/oxytetracycline; vaccines are **serovar-specific bacterins**.
- **Others:** *Borrelia burgdorferi* – Lyme disease, *Ixodes* tick, erythema migrans, lameness in dogs; *B. anserina* – avian spirochaetosis (fowl tick *Argas*). *Treponema pallidum* – syphilis (non-cultivable).

Brachyspira hyodysenteriae — **swine dysentery**, mucohaemorrhagic colitis, strongly β -haemolytic anaerobe.

Mycoplasma & Ureaplasma

SYSTEMATIC

- **Class Mollicutes** — **smallest free-living, self-replicating organisms** (0.2–0.3 μm); **no cell wall** → highly pleomorphic, Gram-unstainable (use Giemsa), **inherently resistant to penicillin & all β -lactams and to lysozyme**; membrane contains **sterol** (needs serum in medium); passes 0.45 μm filters.
- **Culture**: PPLO agar with 20% serum + yeast extract, 5–10% CO_2 , humid; **"fried-egg" colonies** at 3–10 days; **Dienes stain**. *Ureaplasma* = urease +, tiny colonies.
- **Diseases**: *M. mycoides* subsp. *mycoides* **SC** → **CBPP** (contagious bovine pleuropneumonia, marbled lung, notifiable) · *M. capricolum* subsp. *capripneumoniae* → **CCPP** · *M. agalactiae* → contagious agalactia of sheep & goats (mastitis, arthritis, keratoconjunctivitis) · *M. bovis* → mastitis, pneumonia, arthritis, otitis in calves · *M. gallisepticum* → **chronic respiratory disease (CRD) of poultry, egg-transmitted**, detected by rapid serum plate agglutination & HI · *M. synoviae* → infectious synovitis · *M. hyopneumoniae* → enzootic pneumonia of pigs.
- **Treatment**: tylosin, tiamulin, tetracyclines, fluoroquinolones — **never penicillins**.

L-forms vs Mycoplasma: L-forms are cell-wall-deficient variants of ordinary bacteria induced by penicillin/lysozyme and **can revert**; *Mycoplasma* never had a wall and never reverts. Both make fried-egg colonies — the reversion test separates them.

Rickettsiales & Chlamydia

TABLE

Agent	Vector / route	Disease
<i>Anaplasma marginale</i>	Ticks (<i>Rhipicephalus</i> , <i>Hyalomma</i>), fomites, needles	Bovine anaplasmosis — marginal inclusion in RBC on Giemsa, severe anaemia without haemoglobinuria , icterus
<i>Ehrlichia canis</i>	<i>Rhipicephalus sanguineus</i>	Canine monocytic ehrlichiosis — morulae in monocytes, thrombocytopenia, epistaxis ("tracker dog disease")
<i>Coxiella burnetii</i>	Aerosol from birth fluids; ticks; extremely resistant spore-like form	Q fever — abortion in small ruminants; zoonosis, very low infective dose ; phase I/II antigenic variation; survives pasteurisation poorly-done
<i>Neorickettsia risticii</i>	Trematode in aquatic insects	Potomac horse fever — colitis
<i>Rickettsia</i> spp.	Ticks, lice, fleas, mites	Spotted fever & typhus groups; Weil-Felix heterophile test
<i>Chlamydia abortus</i> , <i>C. psittaci</i>	Inhalation, ingestion	Enzootic abortion of ewes; psittacosis/ornithosis in birds & man; <i>C. felis</i> conjunctivitis; <i>C. pecorum</i> polyarthritis

Chlamydial cycle Elementary body = small, rigid, **infectious, metabolically inert** (the "spore") → endocytosis → reorganises into **Reticulate body** = larger, fragile, **non-infectious but dividing** → back to EB → release. Inclusions stained by Giemsa, **Gimenez**, Machiavello. Both *Chlamydia* and *Rickettsia* are **obligate intracellular** — they cannot be grown on cell-free media; use eggs, cell culture or lab animals.

Emerging, re-emerging & transboundary bacteria

RAPID-FIRE

Definitions	Emerging = newly appeared in a population. Re-emerging = previously controlled, now rising. Transboundary (TAD) = highly contagious, crosses borders, major trade impact — WOAHA-listed & notifiable
Drivers	Intensification, live-animal trade, climate & vector range shift, wildlife interface, antimicrobial misuse, urbanisation, ecosystem change
Emerging bacteria	MRSA (LA-MRSA CC398), ESBL/carbapenemase <i>E. coli</i> , <i>Streptococcus suis</i> , <i>Campylobacter jejuni</i> , <i>Yersinia enterocolitica</i> , <i>Borrelia burgdorferi</i> , <i>Coxiella burnetii</i> , <i>Mycoplasma bovis</i>
Re-emerging in India	Anthrax (endemic belts of Odisha, Andhra, Karnataka, Tamil Nadu), brucellosis, bovine TB, glanders in equines, leptospirosis
Transboundary bacterial	CBPP, CCPP, HS, anthrax, glanders, contagious agalactia, <i>Salmonella Gallinarum</i>
Bioterror agents (Cat A)	<i>B. anthracis</i> , <i>Y. pestis</i> , <i>F. tularensis</i> , botulinum toxin
Control framework	One Health ; WOAHA/FAO/WHO tripartite; NADCP & NAP-AMR in India; disease-free zoning & compartmentalisation; risk analysis, surveillance, biosecurity, ring vaccination, stamping out

Two skeletons that cover half the long questions

SKELETON

A · "Write about [any bacterium]" — 8-heading frame

1. Classification & important species
2. Morphology & staining (with the one signature feature)
3. Cultural & biochemical characters (name the selective medium)
4. Antigenic structure / typing scheme
5. Virulence factors & pathogenesis
6. Host, disease, lesions & clinical signs
7. Laboratory diagnosis — sample, direct exam, culture, biochemical, serological, molecular
8. Treatment, immunity, vaccine & control (add the Indian field angle)

B · "Laboratory diagnosis of a bacterial disease" — 6-step frame

1. **Sample:** right tissue, right time, aseptic, with transport medium & cold chain
2. **Direct microscopy:** Gram / ZN / capsule / dark-field / FAT
3. **Isolation:** enrichment → selective & differential media → pure culture; note colony & haemolysis
4. **Identification:** biochemical panel, serotyping, phage typing, MALDI-TOF
5. **Serology:** agglutination, CFT, AGID, ELISA — say what a titre means
6. **Molecular:** PCR/real-time PCR/sequencing; and **ABST by Kirby-Bauer** for the therapeutic answer

Unit 1 · final 90-second sweep

RECALL

Smallest free-living organism	<i>Mycoplasma</i> — no cell wall, sterol membrane	Fried-egg colony	<i>Mycoplasma</i> / L-forms
Dark-field organism	<i>Leptospira</i>	Leptospira reference test	MAT
Leptospira medium	EMJH	CBPP agent	<i>M. mycoides</i> subsp. <i>mycoides</i> SC
CRD of poultry	<i>M. gallisepticum</i>	Infectious form of chlamydia	Elementary body
Q fever agent	<i>Coxiella burnetii</i>	Marginal RBC inclusion	<i>Anaplasma marginale</i>
Essential agent of ovine footrot	<i>Dichelobacter nodosus</i>	Calf diphtheria & liver abscess	<i>Fusobacterium necrophorum</i>
Swine dysentery	<i>Brachyspira hyodysenteriae</i>	Penicillin-resistant by nature	<i>Mycoplasma</i> (no wall)
Obligate intracellular pair	<i>Rickettsia</i> & <i>Chlamydia</i>	Heterophile test for rickettsiae	Weil-Felix

34 / 34 syllabus topics

Paper I · Unit 1

Full pass ≈ 30 min

Night-before pass: sheets 05-06 + all self-tests